Financial Analysis and Decision Making
Global BBA Fall 2026

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Session 6

BUSINESS SCHOOL

SKEMA

Housekeeping

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Plan For Today

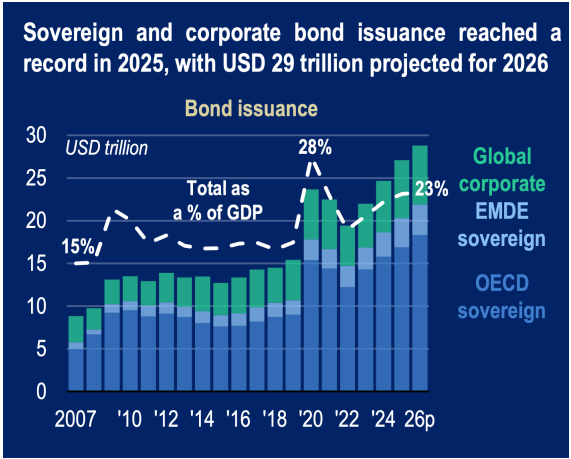


- Bonds
- Introduction to Stocks
- Statistics

Bonds

A bond is a **debt security** in which the issuer/seller (borrower) raises money from investors today in exchange for a promised repayment at the maturity date.

Some Facts: Corporate vs Govt Bonds



Source: Global Debt Report 2026, *OECD*

Who issues bonds?



There are 4 major issuers of bonds.

i. Sovereigns (US, France, Germany, Argentina, etc)

ii. Corporations (Apple, LVMH, Amazon, etc)

iii. Supranational organizations (World Bank, IMF, EU, African Development Bank)

iv. Local governments (but not in France)

Credit Risk



Most bonds are not risk-free, because the issuer may be unable to repay the principal / face value, and default on its debt.

Argentina is particularly famous:

- 2020
- 2014
- 2001
- 1989
- 1982
- ...

Credit Rating Scale

	Moody's	S&P / Fitch	Risk-free +
Investment Grade	Aaa	AAA	0.3%–0.6%
	Aa1–Aa3	AA+/AA/AA-	0.5%–0.9%
	A1–A3	A+/A/A-	0.7%–1.3%
	Baa1–Baa3	BBB+/BBB/BBB-	1.2%–2.2%
Speculative Grade	Ba1–Ba3	BB+/BB/BB-	2.5%–4.5%
	B1–B3	B+/B/B-	4.5%–7%
	Caa1–Caa3	CCC+/CCC/CCC-	7%–12%+
	Ca	CC/C	12%+ (distressed)
	C/D	D	N/A

Global corporate average default rates (1981–2025) (%)

	Time horizon (years)	
Rating	5	10
AAA	0.34	0.68
AA	0.27	0.65
A	0.38	0.99
BBB	1.33	2.78
BB	5.59	10.10
B	15.34	21.67
CCC/C	46.24	50.00
Investment-grade	0.76	1.65
Speculative-grade	13.49	18.87

* Source: S&P Global.

Cost of debt

Suppose Argentina issues a new 10-year zero-coupon bond with a face value of 1000. However, investors are only willing to pay 321.97 for this bond. What is the implied cost of debt / yield on this bond?

Cost of debt

We estimate the cost of debt as

$$R_D = \text{risk-free rate} + \text{credit spread}$$

In most countries, the risk-free rate is given by the yield on the corresponding government bond.

Drivers of the risk-free rate:

- inflation expectations
- health of the economy
- impatience
- investors' demand for risk-free assets

...

Cost of debt

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Drivers of the credit spread:

- default probability
- maturity of the bond
- credit history

...

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What is a stock?

A stock is a certificate of ownership in a company. In contrast to bonds (money that must be repaid), companies sell a percentage of the firm in exchange for equity capital.

Taylor is a penniless CEO of a new start-up. He needs to raise $10m to start his business, but his business does not currently generate revenue, he has no collateral, no credit history, etc.

Taylor convinces his friends to invest $10m in his start-up. He gives them 5 shares of the company, representing 20% ownership. This gives his company a valuation of $50m ($10/20%).

Types of Stocks



There are two major categories of stocks:

Common stock. The most common. Over 99% of stocks are common stock. Usually, 1 share = 1 vote.

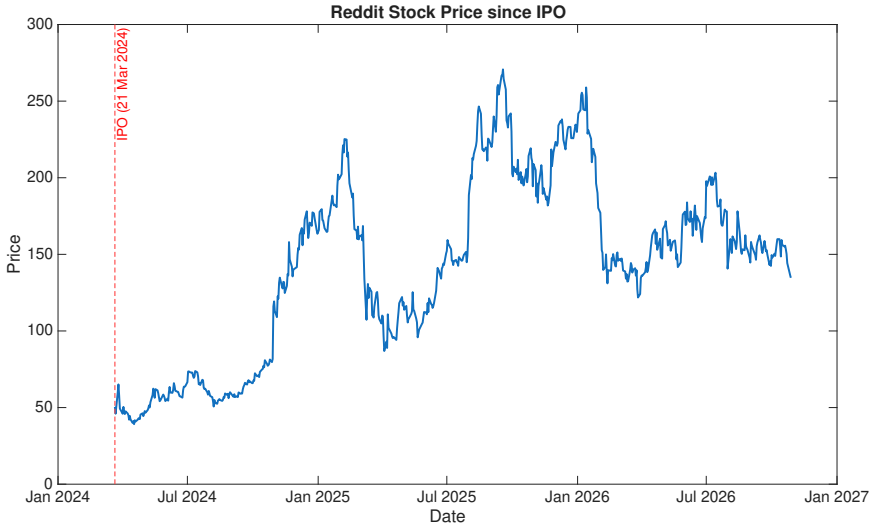
Preferred stock. Equity without voting rights. However, preferred shareholders have priority over common stock in the case of bankruptcy, and are promised regular dividend payments.

Trading stocks on an exchange

Common stock can be bought and sold on exchanges, organized markets where buyers and sellers meet.

- Examples: NYSE and Nasdaq in the US, Euronext Paris in France
- The price is set by supply and demand: more buyers push it up, more sellers push it down
- After an Initial Public Offering (IPO), shares of a company trade among investors.

The case of Reddit



How does buying and selling work?



All buyers on an exchange can post a limit buy order:
“I am willing to buy Apple stock for at most $312.05”.

The **bid** is the highest current offer among buyers.

All sellers on an exchange can post a limit sell order:
“I am willing to sell Apple stock for at least $314.70”.

The **ask** is the lowest current offer among sellers.

How does buying and selling work?



A buyer who places a market/regular buy order (instead of a limit buy order), pays the current ask price.

A seller who places a market/regular sell order, sells at the current bid price.

Example. The current bid on Apple stock is $312.05, and the ask is $313.10. How much does a buyer who places a regular market order pay?

The Bid-Ask Spread

Bid: highest price a buyer will pay.

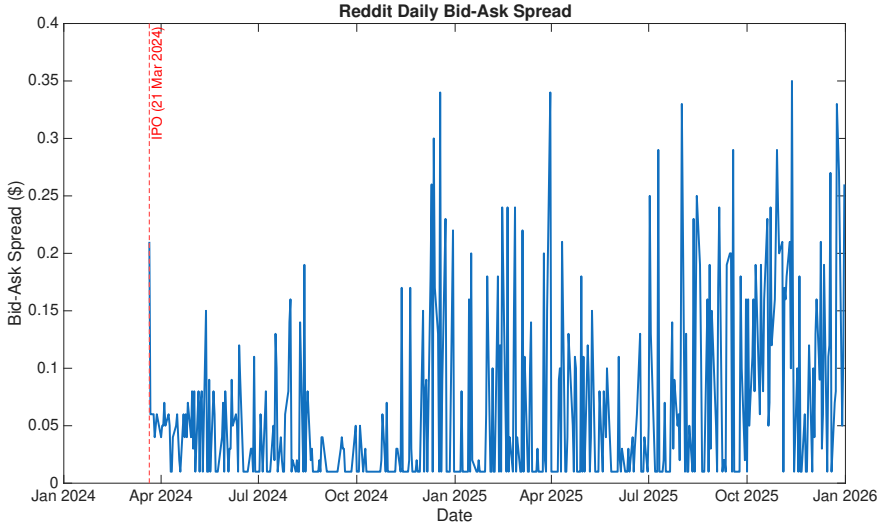
Ask: lowest price a seller will accept.

$$\text{Bid-Ask Spread} = \text{Ask Price} - \text{Bid Price}$$

Example. The current bid on Apple stock is \$312.05, and the ask is \$313.10. What is the current bid-ask spread?

Interpretation. The bid-ask spread is a measure of liquidity: how easily an asset can be bought or sold without too heavily changing the market price.

Bid-Ask Spread for RDDT



Market making

A market maker is an intermediary who facilitates trading by holding inventory of a stock in advance. This avoids the buyer/seller having to wait to find a counterparty to trade.

In exchange for providing liquidity, the market maker can

- Buy at the bid from a potential seller
- Sell at the ask from a potential buyer

The market maker always buys low and sells high!

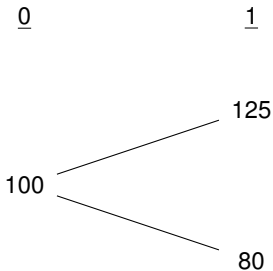
Example. The current bid on Apple stock is $312.05, and the ask is $313.10. What is the market maker's profit per share traded?

Plan For Today



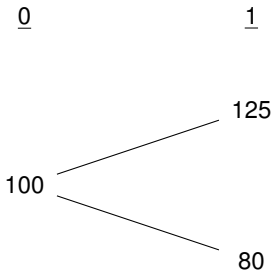
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Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



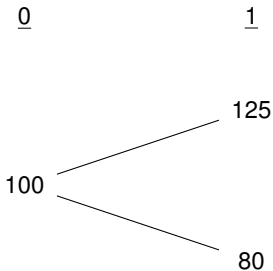
a. If there is equal probability of either state being realized, what is the expected value of the stock in one year?
102.5

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



b. What is the return on this stock at each state?
25% in the good state, -20% in the bad state.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



c. What is the expected return on this stock over the next year?
2.5%

Variance

Variance is a measure of how spread out outcomes are around the expected value, i.e., the average squared surprise.

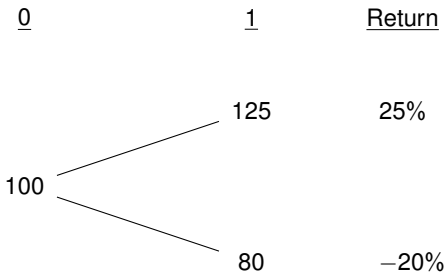
$$\text{Var}(R) = E\left[(R - \mathbb{E}[R])^2\right] = \sum_i \mathrm{p}_i \, (R_i - \mathbb{E}[R])^2$$

where p_i is the probability of each event and $\mathbb{E}[R]$ is the expected value of R.

In our two-state world, the variance is

$$\mathrm{p}_{\text{good}} \times (R_{\text{good}} - \mathbb{E}[R]) + \mathrm{p}_{\text{bad}} \times (R_{\text{bad}} - \mathbb{E}[R])$$

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



d. Calculate the variance of returns.
5.0625%

Volatility

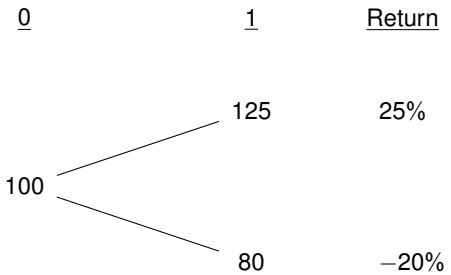
In finance, we define volatility as the standard deviation of a stock's returns, the square-root of the variance,

$$\sigma = \sqrt{\text{Var}(R)} = \sqrt{\sum_i p_i \left(R_i - E[R]\right)^2}$$

Conversely, we write the variance of returns as σ^2.

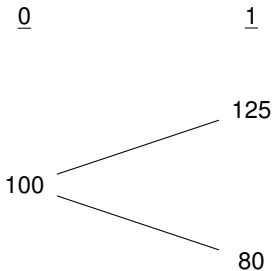
Volatility tells you how much the stock price/return moves around the mean.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



e. Calculate the volatility of returns (Recall, $\sigma^2 = .050625$).
22.5%

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



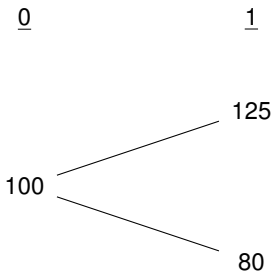
f. For review, re-do parts (a)-(e) but now assuming the probability of the good state is 0.6, and the probability of the bad state is 0.4.

Solutions



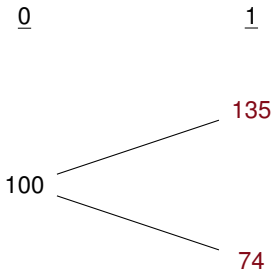
f. For review, re-do parts (a)-(e) but now assuming the probability of the good state is 0.6, and the probability of the bad state is 0.4.
   a. $\mathbb{E}[S_1] = 0.6(125) + 0.4(80) = 75 + 32 = 107$
   b. 25% and -20% as before
   c. $\mathbb{E}[R] = 0.6(0.25) + 0.4(-0.20) = 0.15 - 0.08 = 7\%$
   d. $\text{Var}(R) = 0.6(0.25 - 0.07)^2 + 0.4(-0.20 - 0.07)^2 = 0.0486$
   e. $\sigma = \sqrt{0.0486} = 22.05\%$.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



g. Suppose the probability of the good state is equal to 1. Confirm that the variance is equal to 0.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



h. Confirm that in the new figure above, where the probability of each event equally likely, the volatility is $\sigma = 30.5\%$.

Historical averages and volatility

Suppose we have the following table of past annual returns for a stock.

Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The average return over some fixed horizon is the simple average,

$$\bar{R} = \frac{1}{T}(R_1 + R_2 + \cdots + R_T) = \frac{1}{T}\sum_{t=1}^{T} R_t$$

Historical averages and volatility

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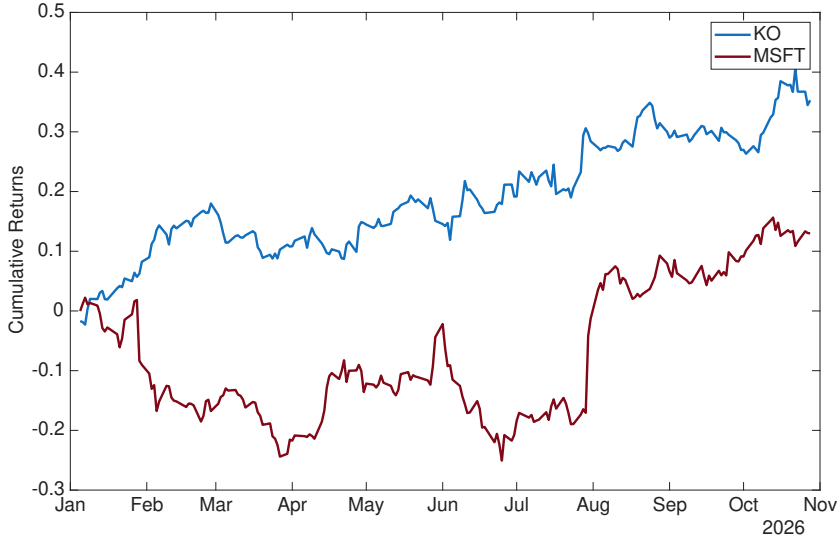
Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The volatility, or population standard deviation of realized returns, measures how much a stock's returns have fluctuated in the past.

$$\bar{\sigma} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(R_t - \bar{R})^2}$$

*This formula is identical to the one we saw before, except now we are treating the probability of each realization being equally representative of the distribution.

Cumulative Returns: KO and MSFT



Cumulative Returns: AAPL and MSFT

